

Technical Internship Report

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Department: Data Analytics Engineering

Project Focus: Data Explainability, Target Isolation, and Machine Learning Leakage Remediation

Reporting Period: June 2, 2026

Task Assignment: Support Demo Preparation (Client Q&A and Data Explainability Framework)

1. Executive Summary

During this training-based internship, my main focus was to learn how data analytics is applied to solve real-world business problems. As part of my learning tasks, I practiced translating technical data engineering concepts into simple, business-friendly explanations.

To practice this, I worked on creating a mock Q&A script and a client demo preparation document. This document was designed to show how our analytical findings (such as flagged claims) can be explained clearly to non-technical teams like account management or customer success. Additionally, I studied data governance concepts, specifically focusing on how to audit data features and prevent data leakage risks in predictive pipelines.

2. Core Technical Tasks & Contributions

Part 1: Client Q&A Support Script (Why Was My Claim Flagged?)

To understand how data findings are shared with non-technical teams, I developed a structured Q&A script as a training assignment. This script translates our data analytical findings into easy-to-understand operational explanations, moving away from automated system assumptions to clear, data-driven insights.

Case 1: High Claim Age Flag

Question 1: "Why is this claim stuck in review for so long?"

- Answer: Through our exploratory data analysis, we evaluated the duration since the medical service was provided (claim_age_days). Based on our analytical criteria, when a claim exceeded acceptable operational thresholds, we flagged it to isolate potential financial backlogs before final submission.

Case 2: Risk-Prone Target Categorization

Question 2: "Why did this specific medical procedure code get flagged?"

- Answer: Our historical data analysis identified patterns where specific procedure codes or insurance types correlated with high payment delays or denials in the past. Consequently, we flagged these risk-prone categories early to fix potential documentation issues.

Case 3: Low Provider Trust / Historical Approval Rating

Question 3: "We submitted everything correctly. Why are we still getting flagged?"

- Answer: We calculated a dynamic historical approval rating for healthcare providers (trust_approval_rate). For providers showing lower recent approval trends in our data, we flagged their current claims to ensure perfect formatting before insurance routing.

Part 2: Data Explainability Framework

I audited and documented the engineered data features within our analytic workflow to understand what variables drive our risk assessment, ensuring our team can explain the data logic transparently.

1. Extracted Predictive Features (Inputs)

- claim_age_days: We used this parameter during feature engineering to calculate the exact number of calendar days a claim has been sitting open. This allows our analysis to capture operational decay and timely filing risks, as older outstanding claims statistically face higher collection friction.
- high_value_flag / repeat_claimant_flag: We engineered these flags to automatically isolate statistical outliers based on extreme billing amounts or structural patterns where identical claims are generated sequentially for the same patient, highlighting clerical duplicate errors.
- seasonal_month: We extracted the calendar month from the transaction log to model and account for seasonal macro-trends. For instance, our data showed that claims processed in December experience unique variations due to year-end patient deductible resets.
- trust_approval_rate: We built this dynamic metric to score healthcare providers based on their historical track record. If a clinic historically maintains a high ratio of approved submissions, our analysis recognizes a stable baseline; if their approval rating drops, we flag new inputs to catch errors early.

2. Operational Review Targets (Outputs)

The model optimizes its risk profiles against core downstream vectors:

- Claim Status & Outcome: The final operational status of the transaction—specifically evaluating whether the claim was fully settled (Paid), permanently Rejected (Denied), or routed back for an in-depth audit.
- Reason Code: The standardized, programmatic error code returned by the payer explaining the exact reason behind an adverse decision or an extended administrative delay (e.g., Authorization not obtained).

- **Follow-up Required & AR Status:** Downstream workflow indicators that show whether a claim is currently trapped in a prolonged open Accounts Receivable balance, automatically signaling if a manual intervention or a customer support action is necessary.

Part 3: Data Leakage Risk Mitigation & Remediation

The Risk Defined:

Data leakage occurs when an algorithm inadvertently trains on information that would not actually be available at the exact moment a prediction must be made. In healthcare billing, metrics like Outcome, Reason Code, Follow-up Required, and AR Status are post-adjudication values—they are determined after a claim has traveled through its lifecycle. If these features are left inside the training dataset, the model appears to have unrealistic, near-perfect accuracy in the lab, but fails completely in production because those data points do not exist yet for a newly submitted claim.

The Remediation Action:

To ensure valid predictive models, I validated the implementation of a rigid Quarantine & Data Separation Isolation Pipeline:

1. **Field Segregation:** High-risk post-adjudication fields (Outcome, Reason Code, Follow-up Required, AR Status) have been cleanly split into a strictly guarded Target/Evaluation zone.
2. **Clean Feature Store Creation:** A dedicated export file (claims_feature_store.csv) is generated containing exclusively pre-adjudication parameters along with safe contextual variables (claim_age_days, trust_approval_rate, etc.).
3. **Elimination of Leakage Vectors:** By dropping quarantine tracking columns completely from the machine learning training inputs, the risk of predictive model contamination is eliminated, allowing the analytical pipeline to rely solely on real-time upfront operational indicators.

3. Key Learning Outcomes & Professional Growth

- **Data Communication:** Learned how to explain technical data concepts (like claim age and approval rates) in simple terms that non-technical people can easily understand.
- **Data Concepts & Cleansing:** Gained a practical understanding of data cleaning, checking for duplicate entries, and the importance of separating data fields to avoid errors (data leakage) in future models.
- **Business Awareness:** Learned how technical data projects are used in real-world business scenarios, such as creating scripts to support product demonstrations for clients.

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